

# TAMILSELVAN G

Full Stack Software Engineer | Angular | .NET Core | AI / RAG | SQL Server | Redis | AWS

+91 7200206323 | [tamilselvang0002@gmail.com](mailto:tamilselvang0002@gmail.com) | Chennai, India | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

## PROFESSIONAL SUMMARY

Full Stack Software Engineer with 4+ years of experience delivering secure, scalable enterprise web applications across the complete SDLC. Expert in Angular (14–20), ASP.NET Core (C#), SQL Server, Docker, and Redis, with measurable impact — 20% faster UI load times, 15% backend efficiency gains, and 99.9% production uptime. Experienced in REST API design, JWT security, SignalR real-time collaboration, Retrieval-Augmented Generation (RAG) and LLM integration (API Key, Ollama), AWS S3, CI/CD automation, Clean Architecture, and Agile/Scrum delivery.

## TECHNICAL SKILLS

**Frontend:** Angular 14–20, TypeScript, JavaScript (ES6+), RxJS, HTML5, CSS3, Tailwind CSS, Bootstrap, Lazy Loading, ApexCharts, TinyMCE, Angular Material, Responsive UI Development, State Management, Reactive Forms

**Backend:** ASP.NET Core, C#, Web API, RESTful Services, SignalR, JWT Authentication, Entity Framework Core, Clean Architecture, SOLID Principles, Middleware, Dependency Injection, Exception Handling

**Database:** SQL Server, Query Optimization, Indexing, Stored Procedures, MongoDB (Basic), Database Design, Performance Optimization

**Testing & Debugging:** Manual Testing, API Testing, Unit Testing, Debugging, Bug Fixing, Postman, Swagger/OpenAPI, Chrome DevTools

**Cache & Infra:** Redis (Caching, Session, Pub/Sub), Docker, Docker Compose, AWS S3, CI/CD, GitHub Actions

**AI & Tools:** Retrieval-Augmented Generation (RAG), LLM Integration (API Key, Ollama), Vector Embeddings, SSE Streaming, Prompt Engineering, Postman, Swagger/OpenAPI, Git, Agile/Scrum

## PROFESSIONAL EXPERIENCE

**Associate Software Engineer (Full Stack) — Nova Techset Pvt. Ltd., Chennai, India**

Jun 2022 – Present

*Progression: Trainee → Junior → Associate Software Engineer*

- Architected and delivered 5+ enterprise web applications using Angular, ASP.NET Core, SQL Server, Docker, and Redis, maintaining 99.9% production uptime.
- Improved backend efficiency by 15% through Redis caching, SQL query optimization, indexing strategies, and architectural refinements.
- Reduced Angular UI load times by 20% via lazy loading, OnPush change detection, RxJS streams, and component-level profiling.
- Designed secure RESTful APIs with JWT authentication and RBAC middleware, hardening security across multiple production applications.
- Implemented real-time collaboration features using SignalR, enabling live multi-user editing and synchronized workflows in publishing platforms.
- Containerized applications using Docker and Docker Compose, standardizing deployments and eliminating environment inconsistencies.
- Integrated AWS S3 with presigned URL workflows for secure, expiry-gated document storage across editorial and vendor platforms.
- Built AI-powered chat assistant using SSE streaming and LLM API integration (Ollama), enabling real-time conversational UX in enterprise apps.
- Established CI/CD pipelines via GitHub Actions, automating build, test, and Docker image deployment workflows.
- Conducted code reviews, built reusable component libraries, and mentored junior developers — reducing post-release defects and improving team delivery quality.

## KEY PROJECTS

### 1. Vendor Management System (VMS) — Publishing Workforce Platform

*Stack: Angular 19, TypeScript, Tailwind CSS, .NET Core 8, C#, SQL Server, Redis, Docker, JWT, REST API, AI Assistant*

- Designed and delivered a full-cycle publishing workforce platform serving 300+ active freelancers — copyeditors, typesetters, proofreaders, and DTP specialists — connecting them with editorial projects through structured onboarding and qualification workflows.

- Built a competency assessment engine for publishing-specific skills with real-time scoring and automated result tracking, reducing vendor vetting time from days to hours.
- Implemented role-based file access control via JWT claims, ensuring manuscript files and editorial assets are accessible only to qualified vendors post-assessment.
- Engineered invoice and billing workflows automating payment calculations based on word counts, deliverable types, and agreed rates — eliminating manual billing overhead.
- Integrated Redis caching for high-frequency job listing and vendor lookup APIs; containerized full stack with Docker for consistent, scalable deployments.
- Integrated AI-powered chat assistant for contextual vendor guidance, reducing onboarding friction and support overhead.

## 2. NEDIT — Enterprise Web-Based Content Editing & XML Proofing Platform

**Stack:** Angular 17, TypeScript, TinyMCE, ICE Plugin, SignalR, .NET Core, SQL Server, XML Processing, REST API

- Contributed to an enterprise-grade content editing and proofing platform enabling end-to-end content management, XML validation, and workflow automation for publishing operations.
- Developed advanced rich-text editing using TinyMCE with text formatting, paragraph styling, footnotes, and dynamic image replacement — enhancing editorial flexibility across structured publishing workflows.
- Built ICE Plugin-based change tracking, enabling accurate revision history, content diffing, and proofing collaboration for distributed editorial teams.
- Implemented real-time multi-user collaboration using SignalR, allowing simultaneous editing with live updates and synchronized workflow state across stakeholders.
- Delivered online mathematical content editing with accurate rendering of complex expressions, and integrated spell-checking and XML validation to ensure error-free, standards-compliant publishing output.
- Ensured XML structural integrity preservation throughout the full proofing lifecycle — from content ingestion through to final publication — maintaining compliance at every workflow stage.

## 3. nPMS — Enterprise Project Management System

**Stack:** Angular 20, .NET Core 10, MongoDB, JWT, REST API, AI Integration

- Own end-to-end development of the frontend, backend, and AI integrations for nPMS, an enterprise project management platform with role-based login and access control across project teams.
- Built an automated issue-tracking and alerting system for project managers handling 100+ issues daily, flagging at-risk or overlooked article issues and notifying PMs in real time — eliminating missed issues and ensuring on-time article delivery.
- Implemented role-based authentication and authorization (RBAC), securing access across project and issue-management modules by user role.

## 4. AI RAG Assistant — Retrieval-Augmented Generation Document Assistant

**Stack:** FastAPI, Python, Qdrant, Redis, Google Gemini API, Ollama, Docker, Render CI/CD

- Designed and built a full-stack RAG (Retrieval-Augmented Generation) document assistant that ingests PDF, DOCX, TXT, and CSV files and answers user questions with citation-grounded, source-referenced responses.
- Engineered the ingestion pipeline — text extraction, recursive chunking (1000 tokens, 200 overlap), and vector embedding generation — storing vectors in Qdrant for high-speed cosine-similarity retrieval (top-k=5).
- Integrated Google Gemini API for production inference — gemini-flash-lite-latest for chat generation and gemini-embedding-001 (768-dimension) for embeddings — with Ollama-based local models (bge-m3, qwen2.5:7b) for local development.
- Used Redis for conversation memory, document registry, and answer caching, reducing latency and redundant LLM calls
- Containerized the application with Docker Compose and deployed it to Render via an automated CI/CD pipeline (live at [ai-rag-agent-rmvy.onrender.com](https://ai-rag-agent-rmvy.onrender.com)).
- Built with an injectable, dependency-based FastAPI service architecture, enabling full unit-test coverage of chunking, embedding, and API layers using mocked external services.

## EDUCATION

**B.E. Computer Science — Alpha Engineering College, Anna University**

Apr 2021

CGPA: 7.45 / 10

## CERTIFICATIONS

- Skill Training Certificate — Web Developer, Government of Tamil Nadu, SCVT (Dec 2021)
- Microsoft Azure Developer Associate (AZ-204) — In Progress